

ECE 3317

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Notes 5

Poynting Theorem



Adapted from notes by Prof. Stuart A. Long

Poynting Theorem

The **Poynting theorem** is one of the most important EM theory. It tells us the power flowing in an electromagnetic field.

John Henry Poynting (1852-1914)



John Henry Poynting was an English physicist. He was a professor of physics at Mason Science College (now the University of Birmingham) from 1880 until his death.

He was the developer and eponym of the Poynting vector, which describes the direction and magnitude of electromagnetic energy flow and is used in the Poynting theorem, a statement about energy conservation for electric and magnetic fields. This work was first published in 1884. He performed a measurement of Newton's gravitational constant by innovative means during 1893. In 1903 he was the first to realize that the Sun's radiation can draw in small particles towards it. This was later coined the Poynting-Robertson effect.

In the year 1884 he analyzed the futures exchange prices of commodities using statistical mathematics.

(Wikipedia)

Some Prerequisites

vector identity $\nabla \cdot (\underline{A} \times \underline{B}) = \underline{B} \cdot (\nabla \times \underline{A}) - \underline{A} \cdot (\nabla \times \underline{B})$

divergence theorem $\int_V \nabla \cdot \underline{A} = \oint_S \underline{A} \cdot \hat{n} dS$

Ohm's law $\underline{J} = \sigma \underline{E}$

Joule's law heat power dissipation $= \int_V \sigma |\underline{E}|^2 dV$

stored electrical energy $\int_V \left(\frac{1}{2} \epsilon |\underline{E}|^2 \right) dV$

stored magnetic energy $\int_V \left(\frac{1}{2} \mu |\underline{H}|^2 \right) dV$

Poynting Theorem

$$\nabla \times \underline{E} = - \frac{\partial \underline{B}}{\partial t}$$

$$\nabla \times \underline{H} = \underline{J} + \frac{\partial \underline{D}}{\partial t}$$

From these we obtain:

$$\underline{H} \cdot (\nabla \times \underline{E}) = - \underline{H} \cdot \frac{\partial \underline{B}}{\partial t}$$

$$\underline{E} \cdot (\nabla \times \underline{H}) = \underline{J} \cdot \underline{E} + \underline{E} \cdot \frac{\partial \underline{D}}{\partial t}$$

Poynting Theorem (cont.)

$$\underline{H} \cdot (\nabla \times \underline{E}) = - \underline{H} \cdot \frac{\partial \underline{B}}{\partial t}$$

$$\underline{E} \cdot (\nabla \times \underline{H}) = \underline{J} \cdot \underline{E} + \underline{E} \cdot \frac{\partial \underline{D}}{\partial t}$$

Subtract, and use the following vector identity:

$$\underline{H} \cdot (\nabla \times \underline{E}) - \underline{E} \cdot (\nabla \times \underline{H}) = \nabla \cdot (\underline{E} \times \underline{H})$$

We then have:

$$\nabla \cdot (\underline{E} \times \underline{H}) = - \underline{J} \cdot \underline{E} - \underline{H} \cdot \frac{\partial \underline{B}}{\partial t} - \underline{E} \cdot \frac{\partial \underline{D}}{\partial t}$$

Poynting Theorem (cont.)

$$\nabla \cdot (\underline{E} \times \underline{H}) = - \underline{J} \cdot \underline{E} - \underline{H} \cdot \frac{\partial \underline{B}}{\partial t} - \underline{E} \cdot \frac{\partial \underline{D}}{\partial t}$$

Next, assume that Ohm's law applies for the electric current:

$$\underline{J} = \sigma \underline{E}$$

→
$$\nabla \cdot (\underline{E} \times \underline{H}) = - \sigma (\underline{E} \cdot \underline{E}) - \underline{H} \cdot \frac{\partial \underline{B}}{\partial t} - \underline{E} \cdot \frac{\partial \underline{D}}{\partial t}$$

or

$$\nabla \cdot (\underline{E} \times \underline{H}) = - \sigma |\underline{E}|^2 - \underline{H} \cdot \frac{\partial \underline{B}}{\partial t} - \underline{E} \cdot \frac{\partial \underline{D}}{\partial t}$$

Poynting Theorem (cont.)

$$\nabla \cdot (\underline{E} \times \underline{H}) = -\sigma |\underline{E}|^2 - \underline{H} \cdot \frac{\partial \underline{B}}{\partial t} - \underline{E} \cdot \frac{\partial \underline{D}}{\partial t}$$

From calculus (chain rule), we have that

$$\underline{E} \cdot \frac{\partial \underline{D}}{\partial t} = \epsilon \left(\underline{E} \cdot \frac{\partial \underline{E}}{\partial t} \right) = \epsilon \frac{1}{2} \frac{\partial}{\partial t} (\underline{E} \cdot \underline{E})$$

$$\underline{H} \cdot \frac{\partial \underline{B}}{\partial t} = \mu \left(\underline{H} \cdot \frac{\partial \underline{H}}{\partial t} \right) = \mu \frac{1}{2} \frac{\partial}{\partial t} (\underline{H} \cdot \underline{H})$$

Hence we have

$$\nabla \cdot (\underline{E} \times \underline{H}) = -\sigma |\underline{E}|^2 - \mu \frac{1}{2} \frac{\partial}{\partial t} (\underline{H} \cdot \underline{H}) - \epsilon \frac{1}{2} \frac{\partial}{\partial t} (\underline{E} \cdot \underline{E})$$

Poynting Theorem (cont.)

$$\nabla \cdot (\underline{E} \times \underline{H}) = -\sigma |\underline{E}|^2 - \mu \frac{1}{2} \frac{\partial}{\partial t} (\underline{H} \cdot \underline{H}) - \epsilon \frac{1}{2} \frac{\partial}{\partial t} (\underline{E} \cdot \underline{E})$$

This may be written as

$$\nabla \cdot (\underline{E} \times \underline{H}) = -\sigma |\underline{E}|^2 - \mu \frac{1}{2} \frac{\partial}{\partial t} |\underline{H}|^2 - \epsilon \frac{1}{2} \frac{\partial}{\partial t} |\underline{E}|^2$$

or

$$\nabla \cdot (\underline{E} \times \underline{H}) = -\sigma |\underline{E}|^2 - \frac{\partial}{\partial t} \left(\frac{1}{2} \mu |\underline{H}|^2 \right) - \frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon |\underline{E}|^2 \right)$$

Poynting Theorem (cont.)

Final differential (point) form of Poynting theorem:

$$\nabla \cdot (\underline{E} \times \underline{H}) = -\sigma |\underline{E}|^2 - \frac{\partial}{\partial t} \left(\frac{1}{2} \mu |\underline{H}|^2 \right) - \frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon |\underline{E}|^2 \right)$$

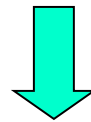
Poynting Theorem (cont.)

Volume (integral) form

Integrate both sides over a volume and then apply the **divergence theorem**:

$$\nabla \cdot (\underline{E} \times \underline{H}) = -\sigma |\underline{E}|^2 - \frac{\partial}{\partial t} \left(\frac{1}{2} \mu |\underline{H}|^2 \right) - \frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon |\underline{E}|^2 \right)$$

$$\int_V \nabla \cdot (\underline{E} \times \underline{H}) dV = - \int_V \sigma |\underline{E}|^2 dV - \int_V \frac{\partial}{\partial t} \left(\frac{1}{2} \mu |\underline{H}|^2 \right) dV - \int_V \frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon |\underline{E}|^2 \right) dV$$



$$\oint_S (\underline{E} \times \underline{H}) \cdot \hat{n} dS = - \int_V \sigma |\underline{E}|^2 dV - \int_V \frac{\partial}{\partial t} \left(\frac{1}{2} \mu |\underline{H}|^2 \right) dV - \int_V \frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon |\underline{E}|^2 \right) dV$$

Poynting Theorem (cont.)

Final **volume** form of Poynting theorem:

$$\oint_S (\underline{E} \times \underline{H}) \cdot \hat{n} dS = - \int_V \sigma |\underline{E}|^2 dV - \int_V \frac{\partial}{\partial t} \left(\frac{1}{2} \mu |\underline{H}|^2 \right) dV - \int_V \frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon |\underline{E}|^2 \right) dV$$

For a **stationary** surface:

$$\oint_S (\underline{E} \times \underline{H}) \cdot \hat{n} dS = - \int_V \sigma |\underline{E}|^2 dV - \frac{\partial}{\partial t} \int_V \left(\frac{1}{2} \mu |\underline{H}|^2 \right) dV - \frac{\partial}{\partial t} \int_V \left(\frac{1}{2} \epsilon |\underline{E}|^2 \right) dV$$

Poynting Theorem (cont.)

Physical interpretation:

(Assume that S is stationary.)

$$-\oint_S (\underline{E} \times \underline{H}) \cdot \hat{n} dS = \int_V \sigma |\underline{E}|^2 dV + \frac{\partial}{\partial t} \int_V \left(\frac{1}{2} \mu |\underline{H}|^2 \right) dV + \frac{\partial}{\partial t} \int_V \left(\frac{1}{2} \epsilon |\underline{E}|^2 \right) dV$$

Power dissipation as heat (Joule's law)

Rate of change of stored magnetic energy

Rate of change of stored electric energy

Right-hand side = power flowing into the region V .

Poynting Theorem (cont.)

Hence

$$- \oint_S (\underline{E} \times \underline{H}) \cdot \hat{n} dS = \text{power flowing into the region}$$

Or, we can say that

$$\oint_S (\underline{E} \times \underline{H}) \cdot \hat{n} dS = \text{power flowing out of the region}$$

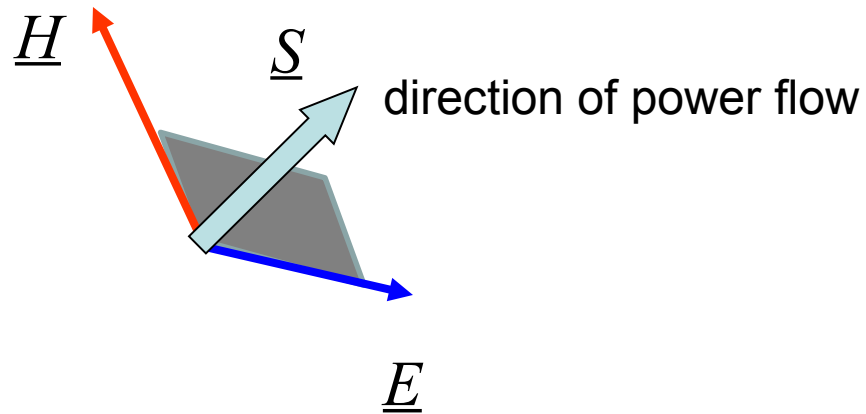
Define the Poynting vector:

$$\underline{S} = \underline{E} \times \underline{H}$$

$$\oint_S \underline{S} \cdot \hat{n} dS = \text{power flowing out of the region}$$

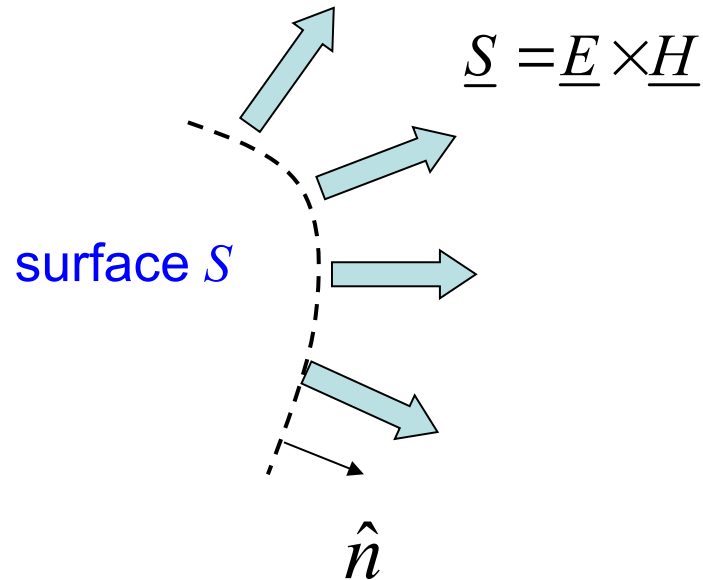
Poynting Theorem (cont.)

$$\underline{S} = \underline{E} \times \underline{H}$$



The units of $\underline{S}$ are [W/m²].

Power Flow



The power P flowing through the surface S (from left to right) is:

$$P = \int_S \underline{S} \cdot \hat{n} dS$$

Time-Average Poynting Vector

Assume sinusoidal (time-harmonic) fields)

$$\underline{S}(x, y, z, t) = \underline{E}(x, y, z, t) \times \underline{H}(x, y, z, t)$$

$$\underline{E}(x, y, z, t) = \text{Re} \left\{ \underline{E}(x, y, z) e^{j\omega t} \right\}$$

$$\underline{H}(x, y, z, t) = \text{Re} \left\{ \underline{H}(x, y, z) e^{j\omega t} \right\}$$

From our previous discussion (notes 2) about time averages, we know that

$$\langle \underline{S}(t) \rangle = \langle \underline{E}(t) \times \underline{H}(t) \rangle = \frac{1}{2} \text{Re}(\underline{E} \times \underline{H}^*)$$

Complex Poynting Vector

Define the **complex Poynting vector**:

$$\underline{S} \equiv \frac{1}{2} (\underline{E} \times \underline{H}^*)$$

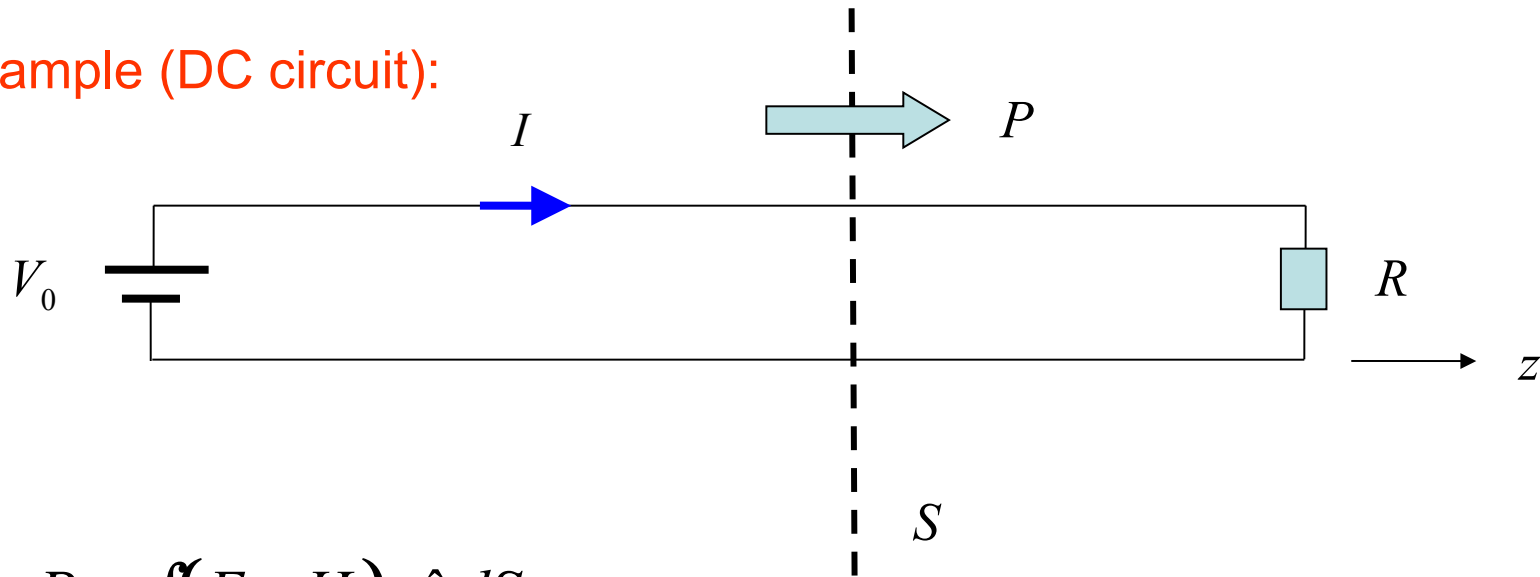
We then have that

$$\langle \underline{S}(x, y, z, t) \rangle = \text{Re}(\underline{S}(x, y, z))$$

Note on Circuit Theory

Although the Poynting vector can always be used to calculate power flow, at low frequency circuit theory can be used, and this is usually easier.

Example (DC circuit):

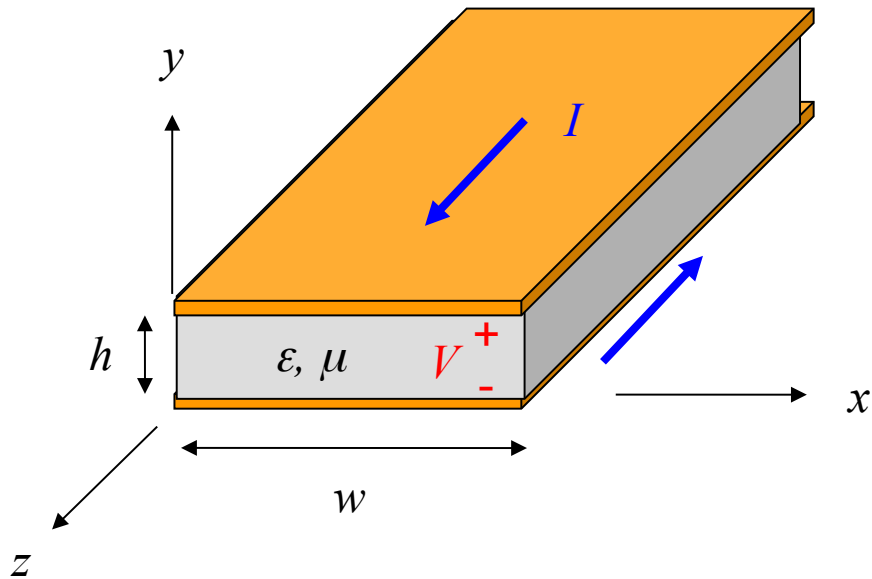


$$P = \int_S (\underline{E} \times \underline{H}) \cdot \underline{\hat{z}} dS$$

$$P = V_0 I \quad \text{if AC, } \langle P \rangle = \frac{1}{2} V_0 I_0^*$$

The second form is much easier to calculate!

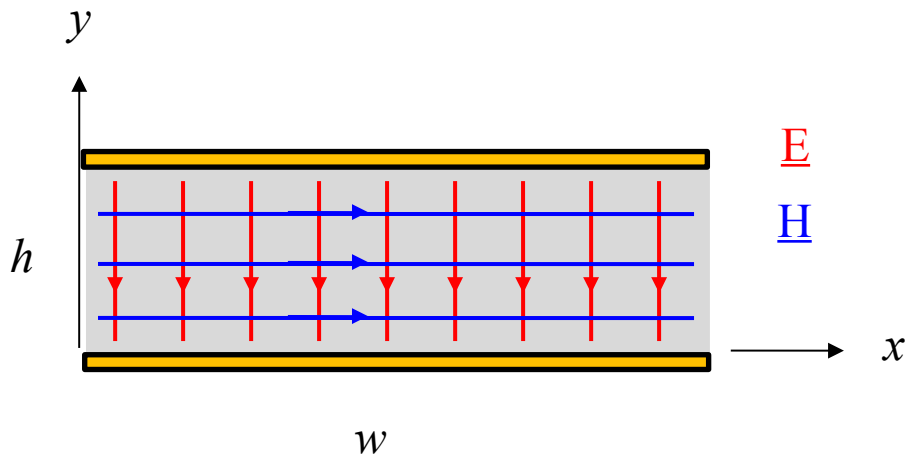
Example: Parallel-Plate Transmission Line



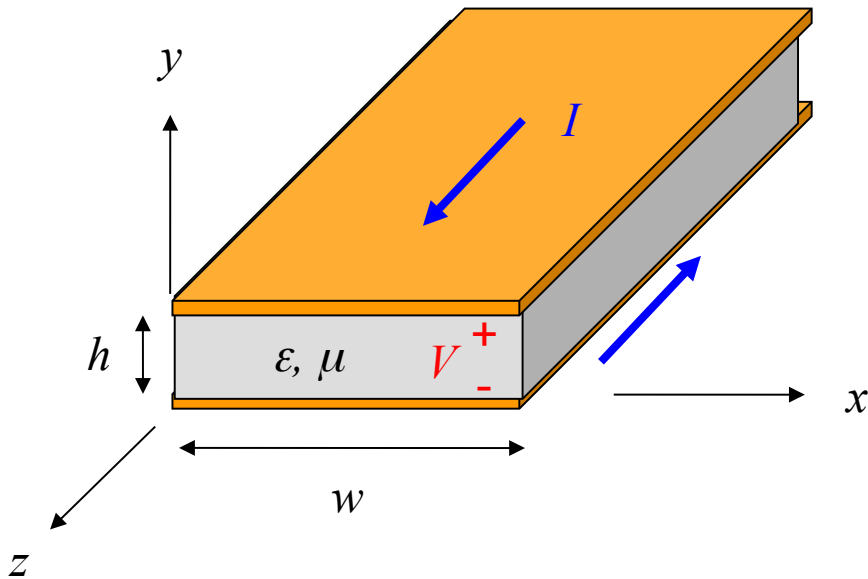
At $z = 0$:

$$V(0) = V_0$$

$$I(0) = I_0$$



Example (cont.)

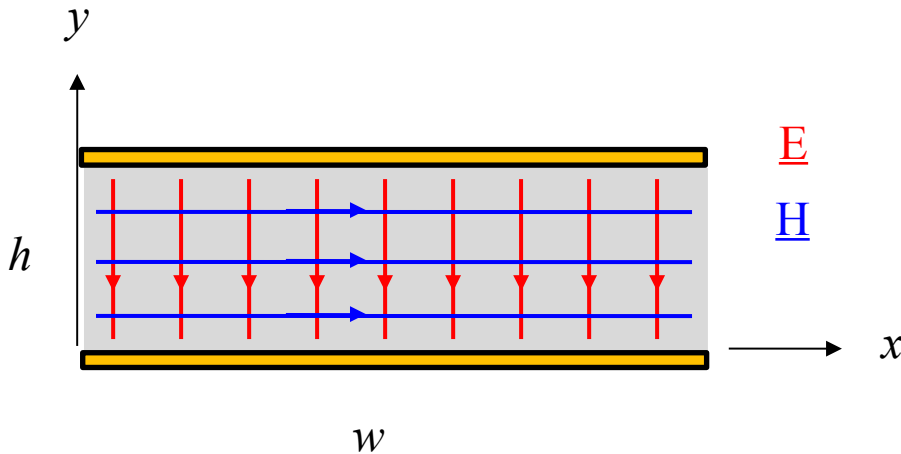


At $z = 0$:

$$\underline{E}(x, y, 0) = -\hat{y} \left(\frac{V_0}{h} \right)$$

$$\underline{H}(x, y, 0) = \hat{x} \left(\frac{I_0}{w} \right)$$

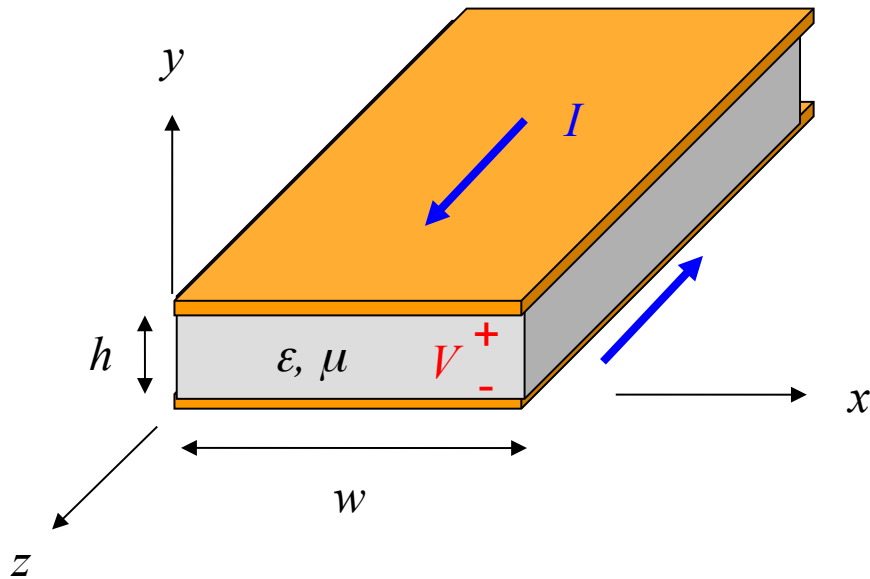
(from ECE 2317/3318)



$$\underline{S} = \frac{1}{2} (\underline{E} \times \underline{H}^*)$$

$$\underline{S} = \frac{1}{2} \hat{z} \left(\frac{V_0}{h} \right) \left(\frac{I_0}{w} \right)^*$$

Example (cont.)

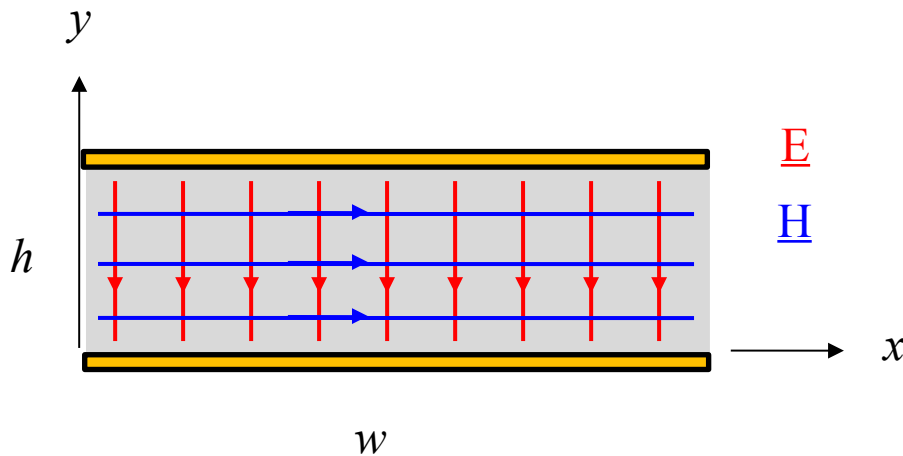


$$\underline{S} = \frac{1}{2} \hat{\underline{z}} \left(\frac{V_0}{h} \right) \left(\frac{I_0}{w} \right)^*$$

Hence

$$P_f = \int_0^h \int_0^w \underline{S} \cdot \hat{\underline{z}} dx dy = \int_0^h \int_0^w S_z dx dy$$

$$P_f = \frac{1}{2} \left(\frac{V_0}{h} \right) \left(\frac{I_0}{w} \right)^* (wh)$$



$$P_f = \frac{1}{2} V_0 I_0^*$$